A search engine tailored for your ISO needs.

ISOOGLE!

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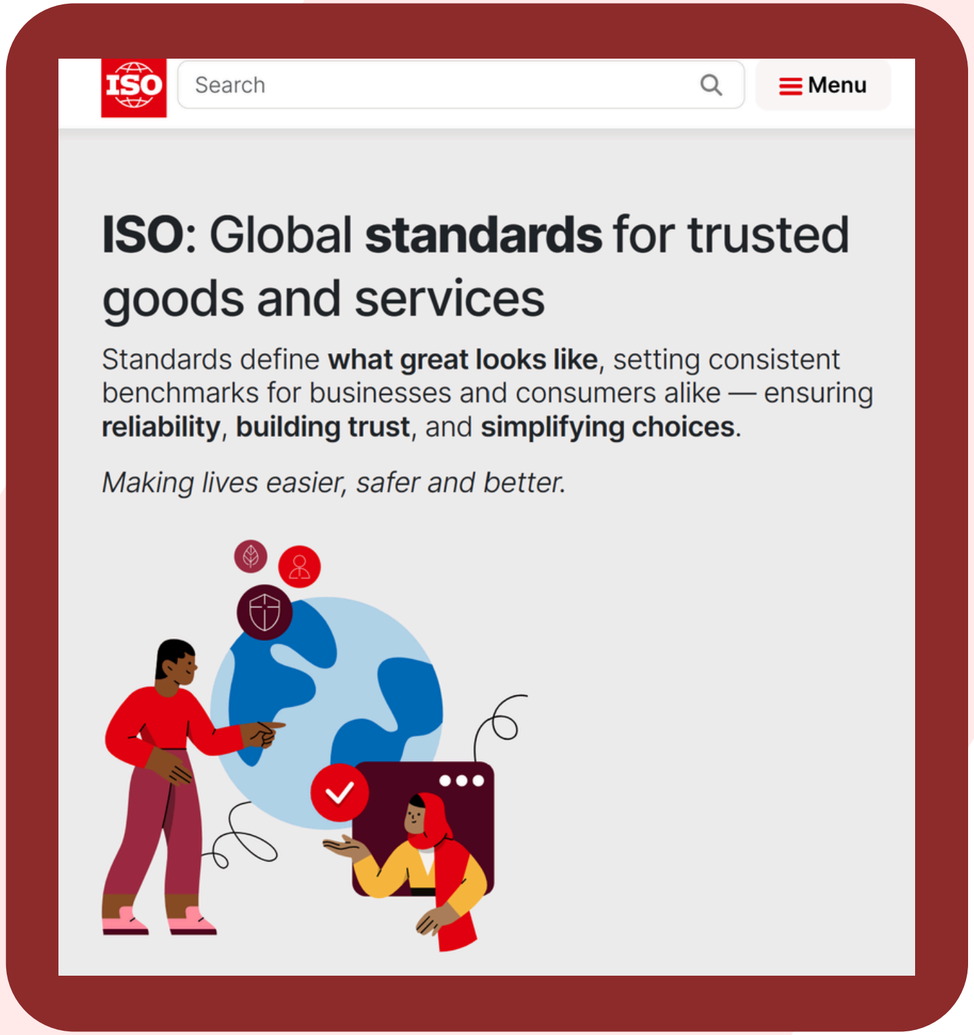
About
ISO

The International Organization for Standardization (ISO) is a non-governmental organization that **develops and publishes standards** to ensure products, services, and systems are safe, reliable, and of high quality.

They are guidelines that span across multiple domains, from information technology to food and agriculture, providing a framework for efficiency and consistency that can **build credibility** and create global market access.



Motivation

The motivation is to **connect disparate applications of ISO standards**.

But ISO is also used **in photography** meaning standardized measurement for film and sensor sensitivity.

A general search engine might struggle to distinguish between the two and retrieve the result a user desires.

Stage 1:
Page Ranking System

1. **Keywords and their Weights**
   - ISO *4
   - Standard *3
   - Sustain *2
   - Certificate *1

2. **Websites**
   - The ISO Official Website
   - The ISO Wikipedia Page
   - CyberArk's Glossary on ISO
   - Eye on Tech – What is ISO
   - ISO Systems UK – Understanding ISO 9001:2015 - Quality Management

3. **Fetch Web content**
   - Fetch and process HTML websites
   - Fetch and process Youtube transcript

4. **Ranking**
   - Rank the pages based on the calculated keyword score

Stage 2:
Site ranking

```
==================================================
                Site Ranking Result
==================================================
Rank #1 [Total Score: 1583]
Site: https://en.wikipedia.org/wiki/International_Organization_for_Standardizati
on
  └─ Main Page Score: 1530 | Counts: {ISO=239, Sustain=5, Standard=188, Certific
ate=0}
  └─ Sub-pages (3):
      ├─ https://en.wikipedia.org/wiki/Main_Page
      │   Score: 28 | Counts: {ISO=6, Sustain=2, Standard=0, Certificate=0}
      ├─ https://en.wikipedia.org/wiki/Wikipedia:Contents
      │   Score: 3 | Counts: {ISO=0, Sustain=0, Standard=1, Certificate=0}
      ├─ https://en.wikipedia.org/wiki/Portal:Current_events
      │   Score: 22 | Counts: {ISO=5, Sustain=1, Standard=0, Certificate=0}
--------------------------------------------------
Rank #2 [Total Score: 710]
Site: https://www.iso.org/home.html
  └─ Main Page Score: 294 | Counts: {ISO=46, Sustain=4, Standard=34, Certificate
=0}
  └─ Sub-pages (3):
      ├─ https://www.iso.org/obp/ui/en/
      │   Score: 0 | Counts: {ISO=0, Sustain=0, Standard=0, Certificate=0}
      ├─ https://www.iso.org/es/home
      │   Score: 201 | Counts: {ISO=42, Sustain=0, Standard=11, Certificate=0}
      ├─ https://www.iso.org/fr/home.html
      │   Score: 215 | Counts: {ISO=47, Sustain=0, Standard=9, Certificate=0}
--------------------------------------------------
Rank #3 [Total Score: 189]
Site: https://www.cyberark.com/what-is/iso/
  └─ Main Page Score: 171 | Counts: {ISO=26, Sustain=0, Standard=15, Certificate
=22}
  └─ Sub-pages (3):
      ├─ https://www.cyberark.com/cdn-cgi/content
      │   Score: 0 | Counts: {}
      ├─ https://www.cyberark.com/events/
      │   Score: 9 | Counts: {ISO=1, Sustain=0, Standard=0, Certificate=5}
      ├─ https://www.cyberark.com/partners/
      │   Score: 9 | Counts: {ISO=1, Sustain=0, Standard=0, Certificate=5}
--------------------------------------------------
```

1

Subpages

- Find 3 child page at most for each main websites (except Youtube video websites)

2

Calculate the site score

- Main page keyword total score *1
- Subpage keyword total score *0.5

site_score =
root_score + child1_score + child2_score + child3_score

3

Rank the whole site

- Rank the sites including child pages

Keyword-Based Domain Filtering

Standards Keywords

Rank	Website	Site Score	Keyword Counts
#1	**Wikipedia ISO Organization**	460	Organization=65, Standard=188, Certification=8, International=92
#2	**Wikipedia Film Speed**	202	Organization=14, Standard=118, Certification=0, International=29
#3	**iso.org**	87	Organization=4, Standard=34, Certification=2, International=15
#4	**Wikipedia Exposure**	23	Organization=4, Standard=6, Certification=0, International=1

Photography Keywords

Rank	Website	Site Score	Keyword Counts
#1	**Wikipedia Exposure**	485.5	Photography=40, Sensitivity=6, Exposure=137, Camera=23
#2	**Wikipedia Film Speed**	335.5	Photography=77, Sensitivity=47, Exposure=115, Camera=86
#3	**Wikipedia ISO Organization**	2	Photography=0, Sensitivity=1, Exposure=0, Camera=1
#4	**iso.org**	0	Photography=0, Sensitivity=0, Exposure=0, Camera=0

Stage 3:
Connect to Google API

Score Range:
0 - 1000+

1. **Google API Querying:**
   - Register Google Custom Search API account
   - Use the user-provided keywords combined with "International Organization of Standardization" to retrieve the top 5 search results

2. **Search Result Preprocessing and Scoring:**
   - Apply data preprocessing procedures and calculate scores for each website with Stage 2 methods

3. **Search Ranking Refinement:**
   - Re-rank the website order based on Stage 2 ranking scores.
   - Return the final ordered list of websites

Stage 4:
Iterative Keyword Search

Score Range:
0 - 1000+

1. **Initial Search:**
   - Original Query Keywords → Weight =10
   - Google API returns Top 5 results

2. **Extract Derived Keywords:**
   - Basic frequency extraction on previous results

   *Stopwords are filtered

3. **Expanded Search:**
   - Execute new search with Expanded Query (Originals + Derived)
   - Derived Keywords → Weight=5
   - ISO - related terms → Weight = 1

4. **Re-score and Rank:**
   - *rank_score* = Σ (count(keyword_original) × 10 +Σ(count(keyword_derived) × 5
     - *node_score* = page_score + Σ (child_node_score)
     - *site_score* = root_node_score

Stage 5:
Semantic Search

Score Range:
0 - 100

1. **Initial Search:**
   - Query Keywords
   - Google API returns results

2. **Compute Keyword Score:**
   - Query keyword → Weight = 4
   - ISO-related terms → Weight =1
   - Inherits formula from Stage 1 + 2

3. **Compute Semantic Score (TF Cosine Similarity):**

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|}$$

*A= Query B=Each page's aggregated content

4. **Normalize, Combine and Rank**
   - Min-Max normalize keyword_score + semantic_score
   - hybridScore=keyword_score*0.6 + semantic_score*0.4

LLM COMPARISON

iso standard

Search Mode / Model	Top 1 URL & Title	Source Type
isoogle Keyword	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
isoogle Iterative	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
isoogle Semantic	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
Perplexity	ISO Standards – iso.org	Official standards body
GPT-4o	ISO Official Website – iso.org	Official standards body

iso camera

Search Mode / Model	Top 1 URL & Title	Source Type
isoogle Keyword	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
isoogle Iterative	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
isoogle Semantic	International Organization for Standardization – Wikipedia	Encyclopedia (Wikipedia)
Perplexity	Camera Basics #5: ISO Sensitivity – Canon Asia	Official brand tutorial (Canon)
GPT-4o	Digital Photography Review (DPReview)	Industry-leading media (DPReview)